Smallest Circle Touching 3 lines in 3D

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This is a simple, yet tedious, problem. Given three lines in 3D (all of infinite length), find the smallest circle that touches all three lines. This is a fairly straightforward algebraic case analysis. Figure 1 lists all examples that need to be considered.

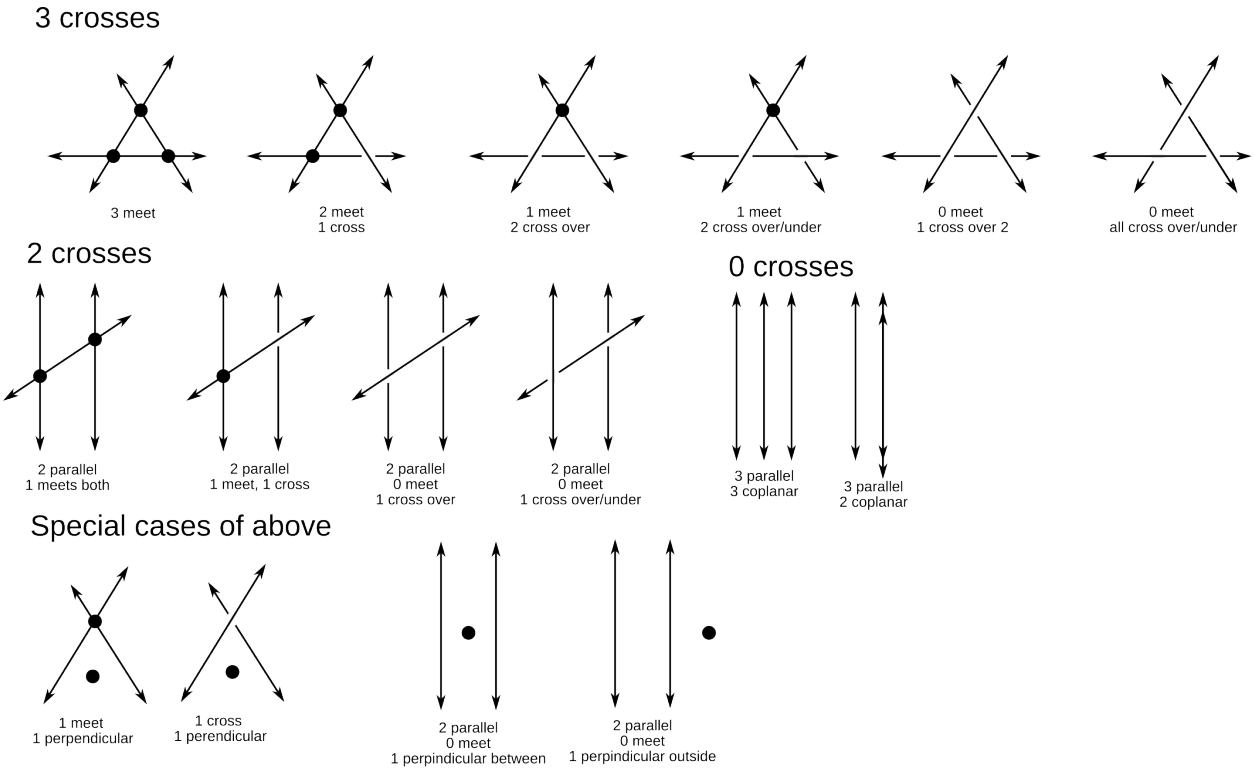


Figure 1: All possible cases of three lines in 3D and how they relate to each other.

The Solution. For any two lines, it is a simple geometric equation to find the closest point between them. Part of this depends on the format the lines are given in (point-slope format vs vector format, etc.). When the lines form a triangle (3 crosses), the circle enscribing the center is the smallest. The goal is, given all three lines in some format, and knowing the case (or being able to quickly determine it), give a closed form solution of the circle (the center point, radius, and the plane it sits on).

Try to find as many solutions as you can!